GAMSA · Gender-Affirming Medicines South Australia

Blood test guide: Oestradiol gel

Oestradiol gel · gamsa.au/monitoring

Key points Take the blood sample before that day's gel, from the arm you did not put gel on. Aim: oestradiol above about 250 pmol/L.

Applies to me	Test	When	Aim	Talk to your prescriber if
☐	**Oestradiol level (gel)**	Before starting, then at each review, after at least 3 to 5 days on the same dose. Take the sample before that day's gel, from the arm you did not put gel on. Rechecked 6 to 8 weeks after a dose change.	Above about 250 pmol/L, with no fixed upper limit Other guidance (2019 position statement): 250 to 600 pmol/L	It is much higher than expected: gel left on the skin where the blood was taken can make the result look far too high
☐	**Total testosterone**	Before starting, then at each review (every 3 to 4 months in the first year, then about once a year)	Below 2 nmol/L, if you want full suppression	It stays above 2 nmol/L when you want full suppression: the blocker or the dose is usually reviewed
☐	**Full blood count, kidney function and electrolytes, liver function**	Before starting, then at each review	Within the range that fits your body on hormones (see Reading your results)	Your prescriber flags a result. Kidney checks matter more if you take spironolactone or PrEP
☐	**LH and FSH**	Before starting. After that, not routine	No fixed aim. LH may be checked to see whether your hormone levels are high enough to protect your bones (this does not work on cyproterone)	—
☐	**Prolactin**	Before starting. After that, only if your prescriber chooses to check it, or if you have symptoms. Best taken in the morning, about 3 to 4 hours after waking, with no hard exercise beforehand	Within the lab's range	You have new headaches, changes in your vision, or a lot of milky nipple discharge
☐	**Lipids and glucose (or HbA1c)**	Before starting, especially with heart risk factors; then periodically. Ask whether you need to fast: most hormone tests do not need it	Within the lab's range	Your prescriber flags a change
☐	**PSA (prostate test)**	Not a routine hormone test. Hormones and genital surgery leave the prostate in place	Read knowing that hormones lower PSA (and finasteride roughly halves it)	Anyone orders a PSA test: tell them you take hormones, and finasteride if you do

A guide to what to expect, not a schedule to follow on your own. Your prescriber sets the timing.

Reading your results

- Lab reports mark results as high or low using the sex on your Medicare or lab record. On hormones, that range may not fit your body. In the first year your prescriber may ask the lab for both the male and the female ranges.
- A single flagged result is a reason to talk with your prescriber, not to change a dose on your own.
- Units: Australian labs report oestradiol in pmol/L, testosterone in nmol/L, and haematocrit as a fraction (0.50 L/L is the same as 50%).

Things that can make a result look wrong

- A sample taken at a different time after the dose from the one your prescriber asked for.
- Oestradiol gel on the skin where the blood is taken: ask for the sample from the arm you did not put gel on. Doing the same with testosterone gel or cream is a sensible precaution, though the guidelines do not say it.
- High-dose biotin (in some hair, skin and nail supplements), which can make some oestradiol tests read falsely high: tell the collection centre if you take it.
- Haemoglobin and haematocrit: read against the range for your affirmed gender once hormones have had time to work (about 3 months on testosterone, about 6 months on oestradiol). On testosterone, the male range is used.
- Kidney function (eGFR) is worked out from your creatinine, age and recorded sex, so the change over time tells more than one result.

Sources: AusPATH 2026 pp21, 31, 32 (Table 5), 36, 42, 54, Table 8 p37; 2019 position statement Baseline, Monitoring; Oestradiol PI s4.4; Endocrine Society 2017 s4.2; Finasteride PI. Full references and detail for clinicians: gamsa.au/monitoring.

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